

SURAFEL DIRIBA

Addis Ababa, Ethiopia

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EDUCATION

Bachelor of Technology in Computer Engineering

Marwadi University, Rajkot, Gujarat, India

Sept 2022 – Feb 2026

GPA: 7.14/10

Scholarship: Study in India Scholarship (Government of India)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Engineering, Discrete Mathematics, Machine Learning

ACADEMIC RESEARCH EXPERIENCE

Undergraduate Research Project – Deepfake Detection Using Eye Blinking Analysis

Supervisor: Prof. Monisha Mohan

May 2025 – Sept 2025

- Conducted research on video-based deepfake detection using physiological eye-blink patterns
- Designed a multi-branch CNN combining dual eye-image streams with Eye Aspect Ratio (EAR) features using depthwise separable convolutions
- Trained and evaluated the model on the FaceForensics++ dataset with parallel video processing and 68-point facial landmark extraction via dlib.
- Research manuscript published in an international journal | [Paper](#) | [GitHub](#)

PROJECTS

OS CPU Scheduling Simulator | [GitHub](#)

Technologies: Java, JavaFx

- Implemented six CPU scheduling algorithms (FCFS, SJF, SRTF, Priority, Round Robin, Exponential Averaging) with real-time visualization
- Designed a process management system using priority queues and linked lists with configurable burst times and context-switch overhead
- Computed and visualized CPU utilization, throughput, turnaround time, and waiting time for comparative analysis

Fake News Detection System | [GitHub](#)

Technologies: Python, scikit-learn, PyTorch, RoBERTa, FastAPI, Docker

- Built an end-to-end ML pipeline for fake news classification using TF-IDF features and speaker credibility metrics
- Integrated Random Forest and transformer-based (RoBERTa) models for real-time inference via FastAPI REST API
- Ensured experiment reproducibility and tracking using Docker and MLflow

CodeTogether: Real-Time Collaborative Editor | [GitHub](#)

Technologies: React, Node.js, Socket.IO

- Developed a real-time collaborative code editor using WebSockets with multi-user synchronization and room-based sessions
- Implemented secure backend architecture for state management and real-time collaboration

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C/C++

CS Concepts: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Machine Learning

Machine Learning: PyTorch, scikit-learn, Transformers (RoBERTa), Feature Engineering, Model Evaluation

Tools & Platforms: Git, Linux, Docker, FastAPI, MongoDB, Firebase